

Operators

def: Operators are special symbols that perform operations on variables & values - math, comparison, or logic.

How to use: placed b/w/around variables $\rightarrow a+b, a==b, a \&\& b$, etc...

when to use: use operators whenever you need to perform a calculation, compare values, combine conditions, or assign/update data.

where to use: Any where you need logic or math - conditions in if statements, loop counters, calculations in expressions:

```
if (age >= 18) { } // relational + logical
```

```
total = price * qty; // arithmetic
```

```
count++;
```

```
// unary, common in loops
```



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why only this way: Java performs integer division when both operands are int - the result stays int, & any system decimal part is truncated (not rounded). This isn't a bug - it reflects how the type system works: an `int/int` operation is defined to produce an `int`. If you want a decimal result, at least one operand must be a `double`:

```
double result = (double) a/b; // 3.333...
```

Real life example: Like tools in a toolbox

hammer (arithmetic) for building, measuring tape (comparison) for checking, combination lock logic (logical) for checking multiple conditions together.